

Problem Statement: US Disaster Declarations Dashboard

Objective

Develop an interactive Power BI dashboard using FEMA disaster declarations data to enable emergency management officials,

policy makers, and analysts to:

- Analyze temporal and geographic patterns of disaster declarations across states and counties.
- Review types of incidents (e.g., Flood, Hurricane, Fire) and program declarations (Individual Assistance, Public Assistance, Hazard Mitigation).
- Assess disaster durations, frequencies, and impacted areas to inform resource allocation and readiness planning.
- Enable ad hoc “What-If” exploration of program resource needs under different disaster frequency scenarios.

Key Requirements

Data Ingestion & Cleaning

- Connect Power BI to us_disaster_declarations.csv (full history) and us_disasters_m5.csv (filtered subset).
- Convert date fields (declaration_date, incident_begin_date, incident_end_date, disaster_closeout_date) to date types; handle missing or malformed dates.
- Standardize categorical fields: state codes, incident_type, declaration_type, and program declaration flags (ia, pa, hm, ih).
- Validate FIPS codes and designated_area values; map numeric FIPS to county names if needed.

Data Modeling

- Construct a star schema with a central “Disaster” fact table and lookup tables for State, IncidentType, DeclarationType, and ProgramTypes.
- Derive calculated fields: DisasterDuration (incident_end_date – incident_begin_date) and TimeToDeclaration (declaration_date – incident_begin_date).
- Aggregate yearly and monthly counts of disasters by state and incident type.

“What-If” Scenario Analysis

- Implement Power BI “What-If” parameters to simulate changes in disaster frequency (e.g., +10%, +20%) and project required public assistance funding based on average per-disaster cost.
- Create DAX measures that recalculate program resource estimates (e.g., total PA funding) under different frequency scenarios.

Visualization & UX

- Page 1: Overview – KPI cards: Total Disasters, Average Duration, Total PA Declarations, Total HM Declarations, Total IA Declarations. Include slicers for year range and incident type.
- Page 2: Geographic Analysis – Choropleth map of states colored by disaster count; bar chart of top 10 counties by disaster count.
- Page 3: Temporal Trends – Line chart of monthly disaster counts; area chart of program declarations over time.
- Page 4: Incident & Program Insights – Clustered bar chart comparing counts by incident type and program; scatter plot of DisasterDuration vs TimeToDeclaration.
- Page 5: Scenario Insights – Line chart of projected PA funding vs frequency adjustment; card showing projected total IA declarations under current “What-If” selection.

Testing & Optimization

- Validate DAX measures against sample manual calculations for 50 random records.
- Ensure date conversions yield correct durations and time differences.
- Optimize data model by hiding unused columns, reducing cardinality in slicers (e.g., group rare incident types into “Other”).
- Ensure dashboard refresh completes within 3 minutes for full dataset (~64,000 rows).

Documentation & Deployment

- User Guide: One-page PDF detailing data refresh, dashboard navigation, and “What-If” parameter usage.
- Technical Document: ERD from Power BI, list of DAX formulas, Power Query M steps for cleaning, instructions for updating dataset when new declarations are published.
- Publish to Power BI Service: Upload .pbix file, configure workspace permissions, set up scheduled refresh (e.g., monthly).